

 AP_040 (/s/profile/0050X000007vw05QAA) (Community Member) asked a question.
June 12, 2019 at 11:11 AM (/s/question/0D50X0000Awa25q\$QA/generated-binary-file-size-in-stm32l4)

Generated binary file size in STM32L4

Is there any specific address location in generated binary file so we can know the size of that binary file?
Using this, in the firmware to read the size of binary file from that location and use it in our application.

STM32 MCUs
(/s/topic/0TO0X000000BSqSWAW/)

STM32L4
(/s/topic/0TO0X000000BWUdWAQ/)

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Top Rated Answers

 Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
2 years ago
If that doesn't work try

```
1      .word _limit_flash - g_pfnVectors /* compute size */
```

 Selected as Best

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All Answers

 Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
2 years ago
One typically has linker symbols, and you can refer to them within the code. For fixed location stuff people typically refer to the symbol via an empty/unused vector in the vector table at the front of the image.
Like Reply

 AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)
2 years ago
@Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
Yes, can you give example for this?
Like Reply

 Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
2 years ago
Not from my phone.
These aren't complicated concepts.
In the GNU tools the symbols are often defined in the linker script, see .LD and .MAP files. In Keil there are \$\$Limit\$\$ type symbols.
Like Reply

 AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)
Edited June 12, 2019 at 12:03 PM
@Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
If some constant value needs to be defined it will be okay but how can I store the binary size? How to calculate the size of .bin file?
Please provide some example to do it?
Which symbol is for Atollic TrueStudio?
Like Reply

 Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
Edited June 12, 2019 at 12:18 PM
I accept PayPal and Amazon Gift cards

<https://community.st.com/s/question/0D70X000006Srs3SAC/end-of-program> (<https://community.st.com/s/question/0D70X000006Srs3SAC/end-of-program>)

```
extern uint32_t Load$$LR$$LR_IROM1$$Limit;  
printf("Limit %08X\n", (uint32_t)&Load$$LR$$LR_IROM1$$Limit);
```

```
extern uint32_t Load$$LR$$LR_IROM1$$Length;  
printf("Length %08X\n", (uint32_t)&Load$$LR$$LR_IROM1$$Length);
```

; In your startup.s file, could be placed at known offset in vector table
IMPORT || Load\$\$LR\$\$LR_IROM1\$\$Length| |
DCD || Load\$\$LR\$\$LR_IROM1\$\$Length| |

In GNU define a variable marking the end of text+code sections

Equivalent to

```
_etext = .; /* define a global symbols at end of code */
```

but including the initialization by placing beyond that.

Like Reply



AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)

2 years ago

@Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)

I tried but it will give errors.

Here I attached the startup and linker file. So, can you suggest me where I can add those symbol to add binary size?



[startup_stm32l462xx](#)
13KB

Like Reply



Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)

2 years ago

Yeah, because that's for Keil, something like this? Perhaps review how your tools function

startup.s

...

g_pfnVectors:

.word _estack

.word Reset_Handler

.word NMI_Handler

.word HardFault_Handler

.word MemManage_Handler

.word BusFault_Handler

.word UsageFault_Handler

.word _limit_flash /* symbol you define in .ld linker script */

.word 0

.word 0

.word 0

.word SVC_Handler

.word DebugMon_Handler

.word 0

...

script.ld

...

/* User_heap_stack section, used to check that there is enough RAM left */

._user_heap_stack :

{

. = ALIGN(4);

PROVIDE (end = .);

PROVIDE (_end = .);

. = . + _Min_Heap_Size;

. = . + _Min_Stack_Size;

. = ALIGN(4);

} >RAM

/* End section */

. = ALIGN(4);

.endof :

{

/* This is used by the startup in order to find the end */

_limit_flash = .; /* define a global symbol at the end of flash */

} >FLASH

/* Remove information from the standard libraries */

/DISCARD/ :

{

libc.a (*)

libm.a (*)

libgcc.a (*)

}

...

Like Reply



AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)

Edited June 12, 2019 at 12:33 PM

For example, binary start at 0x8000000 location and if I read 4bytes from the 0x8000004, i will have the total length of this binary. How I do it?



STM32L462CE_FLASH
4KB

Like Reply

 AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)
2 years ago

Thanks, @Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member)
Now, I am able to get the size of binary from the specific location. But when we load binary into the MCU, it is not working.
So, I put the **".word_limit_flash"** at last in the startup file then location of the size is changed and load .bin file into the MCU, it is working.
Like Reply

 AP_040 (/s/profile/0050X000007vw05QAA) (Community Member)
2 years ago

@Tesla DeLorean (/s/profile/0050X000007vuogQAA) (Community Member) Is there any resource where I can find all those symbol list which we will use in startup file like "_limit_flash"?

Like Reply

More answers

10 of 65

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